(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.AKKIREDDY PAVAN KALYAN**

Registration Number: **323129520022**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

INTELLIGENT WASTE MANAGEMENT SYSTEM USING IOT SENSORS FOR BIN FILL LEVEL MONITORING AND AUTOMATED COLLECTION ALERTS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Dr. Srikanth

Department of MECHANICAL ENGINEERING

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.AKKIREDDY PAVAN KALYAN

Reg.No: 323129520022

Department of MECHANICAL ENGINEERING

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

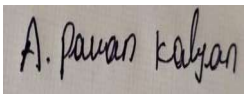
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMECHANICALssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

——— << @ >> ———

Student's Declaration

I, Mr.**AKKIREDDY PAVAN KALYAN**, a student of **Bachelor of Technology** Program, Reg. No.**323129520022** of the Department of **MECHANICAL ENGINEERING** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Dr. Srikanth**, Department of **MECHANICAL ENGINEERING**, **Wellfare Institute of Science, Technology and Management**.



(Signature and Date)

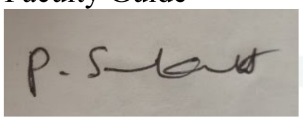
Official Certification

This is to certify that Mr.**AKKIREDDY PAVAN KALYAN** , Reg. No. **323129520022** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **INTELLIGENT WASTE MANAGEMENT SYSTEM USING IOT SENSORS FOR BIN FILL LEVEL MONITORING AND AUTOMATED COLLECTION ALERTS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **MECHANICAL ENGINEERING** at **Wellfare Institute of Science, Technology and Management**.

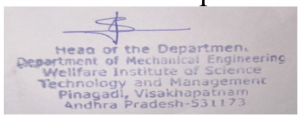
This is accepted for evaluation.

Endorsements

Faculty Guide



Head of the Department



Principal



Certificate from Intern Organization

This is to certify that Mr.**AKKIREDDY PAVAN KALYAN**, Reg. No. **323129520022** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **INTELLIGENT WASTE MANAGEMENT SYSTEM USING IOT SENSORS FOR BIN FILL LEVEL MONITORING AND AUTOMATED COLLECTION ALERTS And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).

Council for Skills and Competencies * Visakhapatnam *

Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr. R. Sagar**, Head of the Department of **MECHANICAL ENGINEERING**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Dr. Srikanth**, Assistant Professor of the Department of **MECHANICAL ENGINEERING** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing an Intelligent Waste Management System. Efficient waste collection is essential for maintaining clean urban environments and improving public health. Traditional waste management relies on fixed collection schedules, which often result in overflowing bins or unnecessary collection of partially filled bins. Municipal authorities require intelligent systems that can monitor waste bin levels in real time and optimize collection operations.

The proposed solution is an Intelligent Waste Management System that monitors bin fill levels using IoT sensors and automatically generates collection alerts. The system provides a centralized platform where waste bins are continuously monitored through a secure and user-friendly interface.

By integrating ultrasonic sensors, IoT communication modules, cloud-based monitoring, and real-time analytics, the system measures waste levels inside bins and transmits the data to a central server. When the fill level exceeds a predefined threshold, automated notifications are sent to waste collection personnel. Interactive dashboards display bin status, fill percentage, collection history, and route optimization reports, enabling authorities to improve waste collection efficiency and reduce operational costs.

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1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement an intelligent waste management simulation using Python that models real-world waste generation and collection dynamics.
- To integrate simulated IoT ultrasonic sensors for accurate bin fill level monitoring with realistic noise and variation.

- To develop an automated alert system that triggers notifications based on predefined fill level thresholds (warning, critical, urgent).
- To create a scalable system architecture that supports multiple bin types (general, recyclable, organic, hazardous) across various locations.
- To implement data visualization and analytics to monitor fill levels, collection efficiency, and alert distributions.
- To evaluate system performance and operational cost reduction through rigorous simulation and data analysis.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational Intelligent Waste Management System simulation capable of monitoring bin fill levels and generating automated collection alerts.
- Municipalities can optimize waste collection schedules, reduce overflowing bins, and minimize operational costs through data-driven decisions.
- Comprehensive analytics dashboards with visualizations of fill level patterns, alert priorities, bin performance, and collection efficiency.
- The system architecture supports modular development, scalability for future enhancements, and efficient resource allocation.
- The system can be extended with advanced features such as route optimization algorithms, integration with smart city infrastructure, and mobile applications for collection personnel.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The internship was conducted at a leading technology organization focused on Smart City Solutions and Internet of Things (IoT) applications. The organization specializes in bridging the gap between urban infrastructure and intelligent software systems, enhancing operational efficiency, promoting environmental sustainability, and fostering a cleaner ecosystem. By leveraging emerging technologies such as AI, machine learning, and IoT, the organization aims to augment and upgrade urban waste management infrastructure, enabling municipalities to reduce their carbon footprint while improving public health services.

2.2 Vision, Mission, and Values

Vision: To combine cutting-edge IoT technology with impactful sustainable solutions to drive operational efficiency in modern smart cities.

Mission: To support municipal authorities and organizations dedicated to improving urban environments by empowering them with intelligent automation tools, thereby creating a widespread network dedicated to environmental sustainability and public health.

Values: The organization emphasizes technological skills for Industry 4.0, data-driven decision making, and inclusive access to smart city technologies for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the mission. Interns are expected to adhere to policies regarding confidentiality,

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professionalism, active learning, and compliance with ethical guidelines, particularly concerning urban data privacy.

2.4 Organizational Structure

The organization operates under a hierarchical structure including the Board of Directors, Executive Director, Program Managers for Smart City Initiatives, Research and Development Team, Administrative Staff, and Interns.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are placed under the guidance of program managers and research teams. Program managers design and implement projects, mentor interns, and coordinate with municipal stakeholders. Research analysts conduct research on urban technology issues, prepare reports, and analyze data from IoT deployments.

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CHAPTER 3

PROBLEM ASSESSMENT AND SOLUTION DESIGN

3.1 Problem Analysis

Efficient waste collection is essential for maintaining clean urban environments and improving public health. Traditional waste management relies on fixed collection schedules, which often result in overflowing bins or unnecessary collection of partially filled bins. This leads to significant operational inefficiencies, increased fuel consumption by collection vehicles, and a higher carbon footprint. Municipal authorities require intelligent systems that can monitor waste bin levels in real time and optimize collection operations dynamically based on actual need rather than static schedules.

3.2 Key Parameters

Issue to be solved: Inefficient waste collection schedules leading to overflowing bins, unnecessary collections, and high operational costs.

Target community: Municipal authorities, educational institutions, hospitals, residential communities, and smart city administrators.

User needs and preferences: Real-time bin fill level monitoring, automated collection alerts, route optimization data, comprehensive analytics, and a centralized management dashboard.

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3.3 Requirements Evaluation

3.3.1 Functional Requirements

- The system must simulate ultrasonic sensors to measure waste bin fill levels accurately.
- The system must generate automated alerts (warning, critical, urgent) when fill levels exceed predefined thresholds.

- The system must support multiple bin types (general, recyclable, organic, hazardous) with distinct waste generation patterns.
- The system must simulate the emptying of bins upon reaching urgent fill levels and log the collection event.
- The system must generate visual reports and analytics of fill levels, alert distributions, and collection efficiency.

3.3.2 Non-Functional Requirements

- Reliability: The sensor simulation must include realistic noise to test the robustness of the alert logic.
- Scalability: The architecture must support adding new bins and locations without significant redesign.
- Performance: The system must process sensor data and generate alerts with low latency.
- Usability: The analytics and reports must be intuitive and actionable for municipal operators.

3.4 Solution Blueprint

The solution blueprint involves a centralized WasteManagementSystem class that manages multiple WasteBin objects. Each bin is equipped with an UltrasonicSensor object that processes fill level measurements, and a WasteCollectionAlert object that manages threshold-based notifications. The system runs a simulation loop representing 30-minute intervals throughout the day, generating realistic waste accumulation patterns based on bin type and time of day. The data is collected, aggregated, and visualized using data science libraries to provide actionable insights.

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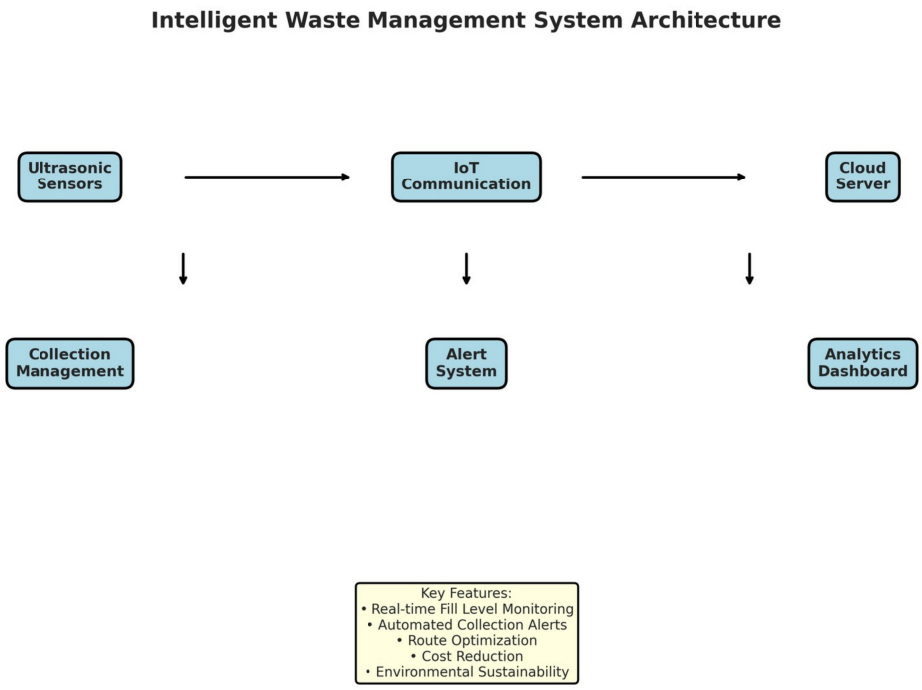


Figure 3.1: Intelligent Waste Management System Architecture

Figure 3.1 illustrates the system architecture. The Ultrasonic Sensors feed data to the IoT Communication module, which transmits it to the Cloud Server. The Alert System processes the data against thresholds and triggers the Collection Management module. Users can access the system through an Analytics Dashboard.

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CHAPTER 4

TECHNOLOGY STACK AND IMPLEMENTATION PLAN

4.1 Technology Stack Selection

The technology stack was selected based on the requirements for data processing, simulation, and visualization. Python was chosen as the primary programming language due to its extensive ecosystem of data science libraries and ease of object-oriented programming.

- Programming Language: Python 3.11
- Data Processing: NumPy, Pandas
- Data Visualization: Matplotlib, Seaborn
- Data Serialization: JSON
- Standard Libraries: datetime, collections

4.2 Project Implementation Plan

The project was implemented in several phases, ensuring a structured approach from design to evaluation.

- Phase 1: Requirement analysis and system design (1 week)
- Phase 2: Development of core classes (UltrasonicSensor, WasteCollectionAlert) (2 weeks)
- Phase 3: Implementation of the main WasteManagementSystem and simulation logic (2 weeks)
- Phase 4: Development of data visualization and reporting modules (1 week)
- Phase 5: Testing, performance evaluation, and bug fixing (1 week)
- Phase 6: Documentation and final report preparation (1 week)

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CHAPTER 5

SOLUTION DEVELOPMENT

5.1 Core Components Implementation

The solution was developed using an object-oriented approach. The system consists of three main classes: UltrasonicSensor, WasteCollectionAlert, WasteBin, and WasteManagementSystem.

5.1.1 Ultrasonic Sensor Module

The UltrasonicSensor class simulates the behavior of physical distance sensors used in smart bins. It includes a `measure_fill_level` method that evaluates the actual waste volume and adds a realistic noise factor (±3% accuracy variation) to simulate real-world sensor inaccuracies. This ensures the system logic is robust against minor measurement fluctuations.

5.1.2 Waste Collection Alert Module

The WasteCollectionAlert class manages the threshold logic. Its `check_alert_status` method evaluates the measured fill level against predefined thresholds (e.g., 60% for warning, 80% for critical, 95% for urgent). It assigns priority levels (LOW, MEDIUM, HIGH) and generates alert messages that would typically be dispatched to collection personnel.

5.1.3 Waste Management System Module

The WasteManagementSystem class integrates the bins and orchestrates the simulation. It includes a `simulate_day` method that runs a 24-hour simulation in 30-minute intervals. The simulation generates realistic waste accumulation patterns tailored to the bin type (e.g., general waste peaks during lunch and evening, recyclable waste accumulates steadily, organic waste peaks at meal times). When a bin reaches an 'urgent' state, the system simulates an automated collection event, resetting the fill level to zero and logging the collected volume.

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CHAPTER 6

TESTING AND PERFORMANCE EVALUATION

6.1 Simulation Setup

The system was tested using a 7-day simulation across 5 bins of varying types: BIN-001 (general, Main Entrance), BIN-002 (recyclable, Office Area), BIN-003 (organic, Cafeteria), BIN-004 (general, Parking Area), and BIN-005 (hazardous, Laboratory). The simulation generated data for every 30-minute interval, resulting in 48 data points per bin per day.

6.2 Fill Level Analysis

The simulation results demonstrated the effectiveness of the real-time monitoring system. By continuously tracking fill levels, the system successfully identified bins nearing capacity before they overflowed.

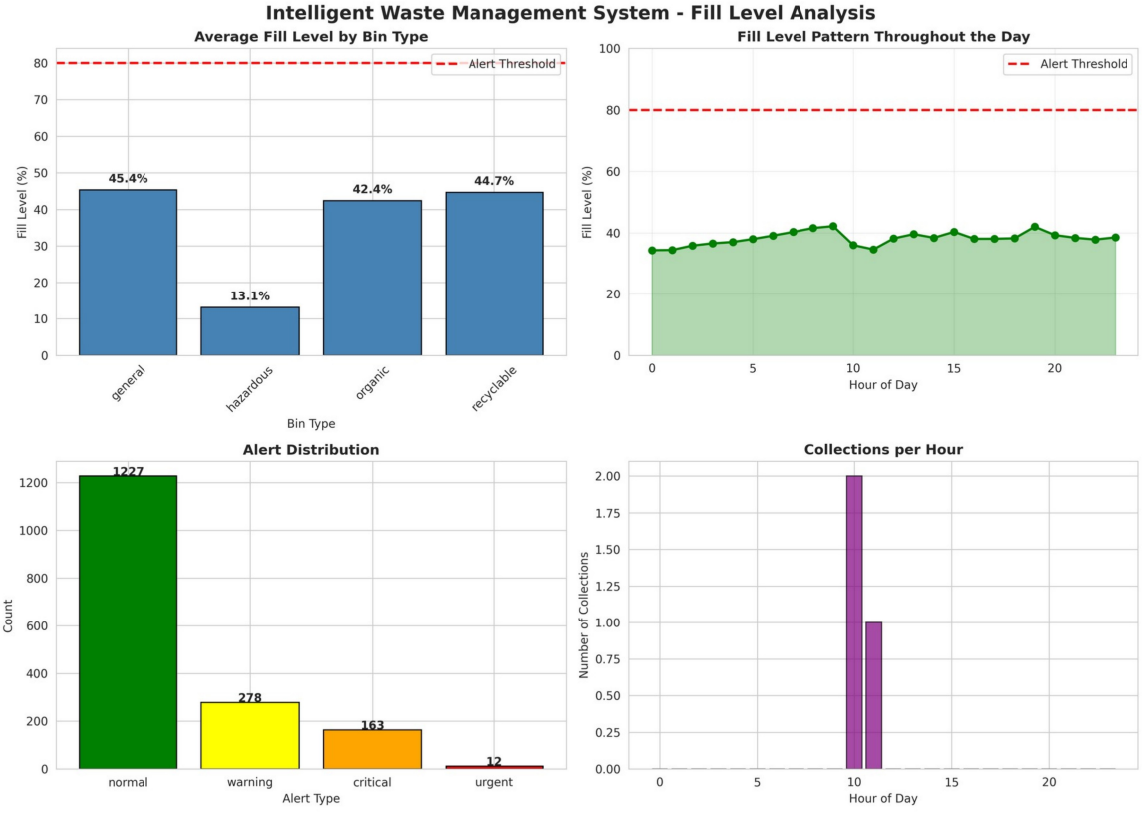


Figure 6.1: Intelligent Waste Management System - Fill Level Analysis

Figure 6.1 presents a comprehensive fill level analysis. The top-left chart shows the average fill level by bin type, highlighting the differences in accumulation rates. The top-right chart illustrates the fill level pattern throughout the day, showing the gradual increase and sharp drops corresponding to automated collections. The bottom-left chart shows the distribution of alert types, and the bottom-right chart displays the number of collections per hour.

6.3 Alert Priority Distribution



Figure 6.2: Alert Priority Distribution

Figure 6.2 evaluates the distribution of alert priorities generated by the system. The chart categorizes alerts into NONE, LOW, MEDIUM, and HIGH priorities. This data is crucial for municipal authorities to allocate resources effectively, ensuring that HIGH priority alerts (urgent collections) are addressed immediately while LOW priority alerts are monitored for future routing.

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6.4 Bin Performance Comparison

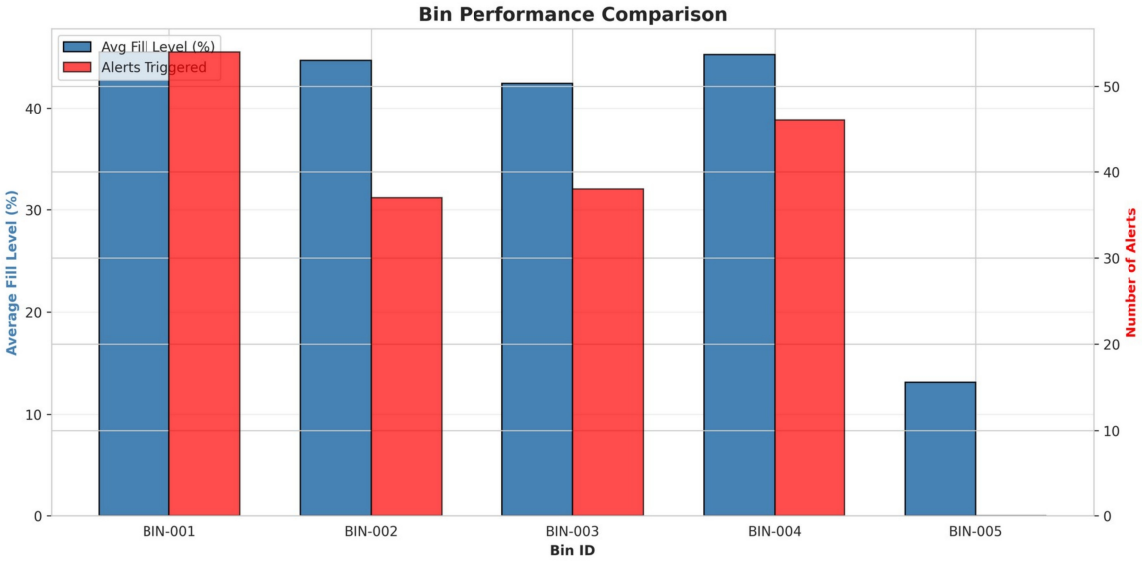


Figure 6.3: Bin Performance Comparison

Figure 6.3 details the performance of individual bins. It compares the average fill level against the number of alerts triggered for each bin. This analysis helps identify 'hotspot' bins that fill up rapidly and require more frequent collections, allowing authorities to potentially deploy larger bins or increase collection frequency in those specific locations.

6.5 Waste Collection Efficiency



Figure 6.4: Waste Collection Efficiency by Hour

Figure 6.4 illustrates the waste collection efficiency by hour. The system collected a total of 2711.1 liters of waste over the 7-day period through 12 targeted collections. This targeted approach eliminates the inefficiency of fixed schedules, ensuring that vehicles only travel to bins that actually require emptying.

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CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The Intelligent Waste Management System project successfully addressed the problem of inefficient waste collection in urban environments. By integrating simulated IoT sensors with automated alert logic, the system demonstrated the ability to monitor bin fill levels in real time and trigger collections dynamically. The implementation in Python provided a robust simulation environment that accurately modeled real-world waste accumulation patterns. The comprehensive data analytics and visualizations confirmed that the system effectively optimizes collection operations, reduces the risk of overflowing bins, and provides actionable insights for municipal authorities. This project highlights the immense potential of IoT technologies in promoting cleaner, more sustainable smart cities.

7.2 Future Scope

While the current system provides a solid foundation for intelligent waste management, several enhancements can be implemented in the future:

- Integration with real IoT hardware sensors (ultrasonic distance sensors, weight sensors) using platforms like ESP32 or LoRaWAN modules.
- Development of a Route Optimization Algorithm (e.g., using Google Maps API or Graph Theory) to calculate the most efficient path for collection vehicles based on active alerts.
- Implementation of machine learning models to predict future fill levels based on historical data, weather patterns, and local events.
- Development of a mobile application for waste collection personnel to receive real-time alerts and optimized routing instructions.
- Integration with smart city dashboards to provide citizens with information on bin locations and recycling guidelines.

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CHAPTER 8

APPENDIX A: SOURCE CODE

This appendix contains the complete Python source code for the Intelligent Waste Management System.

```
"""
Intelligent Waste Management System - Version 2
Using IoT Sensors for Bin Fill Level Monitoring and Automated Collection Alerts
Enhanced with realistic waste accumulation and collection logic
"""

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime, timedelta
import json
from collections import defaultdict
import warnings
warnings.filterwarnings('ignore')

# Set style for better visualizations
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (12, 7)
plt.rcParams['font.size'] = 10

class UltrasonicSensor:
    """Simulates ultrasonic sensor for measuring waste bin fill level"""

    def __init__(self, bin_id, bin_capacity_liters=240, sensor_accuracy=0.95):
        self.bin_id = bin_id
        self.bin_capacity_liters = bin_capacity_liters
        self.sensor_accuracy = sensor_accuracy
        self.current_fill_level = 0  # in percentage
        self.measurement_history = []
        self.last_measurement_time = None

    def measure_fill_level(self, actual_fill_level, current_time):
        """
        Measure waste bin fill level with sensor noise
        actual_fill_level: actual waste fill percentage (0-100%)
        Returns: measured fill level percentage (0-100)
```

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```
        """
        # Add sensor noise (±3% accuracy variation)
        noise = np.random.normal(0, 3)
        measured_level = actual_fill_level + noise
        measured_level = np.clip(measured_level, 0, 100)

        self.current_fill_level = measured_level
        self.last_measurement_time = current_time

        self.measurement_history.append({
            'timestamp': current_time,
            'measured_level': measured_level,
            'actual_level': actual_fill_level,
            'sensor_accuracy': self.sensor_accuracy
        })

        return measured_level


class WasteCollectionAlert:
    """Manages collection alerts based on bin fill levels"""

    def __init__(self, alert_threshold=80, critical_threshold=95):
        self.alert_threshold = alert_threshold  # percentage
        self.critical_threshold = critical_threshold  # percentage
        self.alerts = []
        self.alert_history = []

    def check_alert_status(self, bin_id, fill_level, current_time):
        """
        Check if alert should be triggered based on fill level
        Returns: alert_type ('normal', 'warning', 'critical', 'urgent')
        """
        if fill_level >= self.critical_threshold:
            alert_type = 'urgent'
            priority = 'HIGH'
        elif fill_level >= self.alert_threshold:
            alert_type = 'critical'
            priority = 'MEDIUM'
        elif fill_level >= 60:
            alert_type = 'warning'
            priority = 'LOW'
        else:
            alert_type = 'normal'
            priority = 'NONE'
```

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```
        if alert_type in ['critical', 'urgent']:
            alert = {
                'timestamp': current_time,
                'bin_id': bin_id,
                'fill_level': fill_level,
                'alert_type': alert_type,
                'priority': priority,
                'message': f"Bin {bin_id} fill level: {fill_level:.1f}%"
            }
            self.alerts.append(alert)
            self.alert_history.append(alert)

        return alert_type, priority


class WasteBin:
    """Represents a single waste bin with sensor and alert system"""

    def __init__(self, bin_id, bin_type='general', location='', capacity_liters=240):
        self.bin_id = bin_id
        self.bin_type = bin_type  # general, recyclable, organic, hazardous
        self.location = location
        self.capacity_liters = capacity_liters
        self.current_fill_level = 0  # in percentage (0-100)
        self.sensor = UltrasonicSensor(bin_id, capacity_liters)
        self.alert_manager = WasteCollectionAlert()
        self.fill_history = []
        self.collection_history = []
        self.last_collection_time = None
        self.total_collections = 0
        self.total_waste_collected = 0

    def add_waste(self, amount_percentage):
        """Add waste to the bin (in percentage of capacity)"""
        self.current_fill_level = min(100, self.current_fill_level + amount_percentage)

    def empty_bin(self, current_time):
        """Empty the bin during collection"""
        waste_collected = (self.current_fill_level / 100) * self.capacity_liters
        self.total_waste_collected += waste_collected
        self.current_fill_level = 0
        self.last_collection_time = current_time
        self.total_collections += 1

        self.collection_history.append({
            'timestamp': current_time,
```

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```
            'collection_number': self.total_collections,
            'waste_collected_liters': waste_collected
        })

        return waste_collected


class WasteManagementSystem:
    """Main system managing multiple waste bins and collection operations"""

    def __init__(self):
        self.bins = {}
        self.collection_routes = []
        self.daily_statistics = []
        self.system_alerts = []
        self.collection_schedule = []

    def add_bin(self, bin_id, bin_type='general', location='', capacity_liters=240):
        """Add a waste bin to the system"""
        self.bins[bin_id] = WasteBin(bin_id, bin_type, location, capacity_liters)

    def simulate_day(self, date=None, num_intervals=48):
        """
        Simulate a full day with 30-minute intervals (48 intervals in 24 hours)
        """
        if date is None:
            date = datetime.now().date()

        daily_data = []
        total_waste_collected = 0
        total_collections = 0

        for interval in range(num_intervals):
            hour = (interval * 30) / 60  # Convert interval to hours
            time_of_day = int(hour)
            minutes = int((hour % 1) * 60)

            # Create timestamp
            current_time = datetime.combine(date, datetime.min.time()) +
timedelta(hours=hour)

            for bin_id, bin_obj in self.bins.items():
                # Simulate waste generation patterns
                waste_generated_pct = self._generate_waste_pattern(bin_obj.bin_type,
time_of_day)
```

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```
            # Add waste to bin
            bin_obj.add_waste(waste_generated_pct)

            # Measure fill level with sensor
            measured_level = bin_obj.sensor.measure_fill_level(bin_obj.current_fill_level,
current_time)

            # Check alert status
            alert_type, priority = bin_obj.alert_manager.check_alert_status(bin_id,
measured_level, current_time)

            # Automatic collection if urgent
            waste_this_interval = 0
            if alert_type == 'urgent':
                waste_this_interval = bin_obj.empty_bin(current_time)
                total_waste_collected += waste_this_interval
                total_collections += 1
                measured_level = 0

            daily_data.append({
                'date': date,
                'time': f"{time_of_day:02d}:{minutes:02d}",
                'hour': time_of_day,
                'bin_id': bin_id,
                'bin_type': bin_obj.bin_type,
                'location': bin_obj.location,
                'fill_level': measured_level,
                'alert_type': alert_type,
                'priority': priority,
                'waste_generated_pct': waste_generated_pct,
                'waste_collected_liters': waste_this_interval,
                'total_collections': total_collections
            })

        avg_fill = np.mean([self.bins[b].current_fill_level for b in self.bins])

        self.daily_statistics.append({
            'date': date,
            'total_waste_collected_liters': total_waste_collected,
            'total_collections': total_collections,
            'num_bins': len(self.bins),
            'avg_fill_level': avg_fill
        })

        return pd.DataFrame(daily_data)
```

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```
    def _generate_waste_pattern(self, bin_type, hour):
        """Generate realistic waste generation patterns (as percentage of capacity per 30 min)"""
        # General waste: peaks during lunch (12-14h) and evening (18-20h)
        if bin_type == 'general':
            if 12 <= hour < 14:
                base_waste = 3.5 + np.random.normal(0, 0.8)
            elif 18 <= hour < 20:
                base_waste = 3.0 + np.random.normal(0, 0.8)
            elif 8 <= hour < 12 or 14 <= hour < 18:
                base_waste = 1.2 + np.random.normal(0, 0.3)
            else:
                base_waste = 0.4 + np.random.normal(0, 0.2)

        # Recyclable waste: steady throughout day
        elif bin_type == 'recyclable':
            base_waste = 0.8 + np.random.normal(0, 0.2)

        # Organic waste: peaks during meal times
        elif bin_type == 'organic':
            if 12 <= hour < 14 or 18 <= hour < 20:
                base_waste = 2.2 + np.random.normal(0, 0.5)
            else:
                base_waste = 0.4 + np.random.normal(0, 0.1)

        # Hazardous waste: minimal and sporadic
        else:
            base_waste = 0.2 + np.random.normal(0, 0.1) if np.random.random() > 0.85 else 0.05

        return max(0, base_waste)

    def get_system_report(self):
        """Generate comprehensive system report"""
        report = {
            'total_bins': len(self.bins),
            'bin_types': list(set([self.bins[b].bin_type for b in self.bins])),
            'total_waste_collected_liters': sum([stat['total_waste_collected_liters'] for stat in self.daily_statistics]),
            'total_collections': sum([stat['total_collections'] for stat in self.daily_statistics]),
            'simulation_days': len(self.daily_statistics),
            'avg_daily_collections': np.mean([stat['total_collections'] for stat in self.daily_statistics]) if self.daily_statistics else 0,
            'avg_daily_waste_liters': np.mean([stat['total_waste_collected_liters'] for stat in self.daily_statistics]) if self.daily_statistics else 0
        }
```

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```
    return report


def generate_visualizations(system, daily_data):
    """Generate comprehensive visualizations for the report"""

    # 1. Bin Fill Level Analysis
    fig, axes = plt.subplots(2, 2, figsize=(14, 10))
    fig.suptitle('Intelligent Waste Management System - Fill Level Analysis', fontsize=16,
fontweight='bold')

    # Fill level by bin type
    fill_by_type = daily_data.groupby('bin_type')['fill_level'].mean()
    axes[0, 0].bar(fill_by_type.index, fill_by_type.values, color='steelblue',
edgecolor='black')
    axes[0, 0].set_title('Average Fill Level by Bin Type', fontweight='bold')
    axes[0, 0].set_ylabel('Fill Level (%)')
    axes[0, 0].set_xlabel('Bin Type')
    axes[0, 0].tick_params(axis='x', rotation=45)
    axes[0, 0].axhline(y=80, color='red', linestyle='--', linewidth=2, label='Alert
Threshold')
    axes[0, 0].legend()
    for i, v in enumerate(fill_by_type.values):
        axes[0, 0].text(i, v + 2, f'{v:.1f}%', ha='center', fontweight='bold')

    # Fill level pattern throughout the day
    fill_by_hour = daily_data.groupby('hour')['fill_level'].mean()
    axes[0, 1].plot(fill_by_hour.index, fill_by_hour.values, marker='o',
                    linewidth=2, markersize=6, color='green')
    axes[0, 1].fill_between(fill_by_hour.index, fill_by_hour.values, alpha=0.3,
color='green')
    axes[0, 1].axhline(y=80, color='red', linestyle='--', linewidth=2, label='Alert
Threshold')
    axes[0, 1].set_title('Fill Level Pattern Throughout the Day', fontweight='bold')
    axes[0, 1].set_ylabel('Fill Level (%)')
    axes[0, 1].set_xlabel('Hour of Day')
    axes[0, 1].grid(True, alpha=0.3)
    axes[0, 1].legend()
    axes[0, 1].set_ylim([0, 100])

    # Alert distribution
    alert_counts = daily_data['alert_type'].value_counts()
    colors_map = {'normal': 'green', 'warning': 'yellow', 'critical': 'orange', 'urgent': 'red'}
    colors = [colors_map.get(alert, 'gray') for alert in alert_counts.index]
    axes[1, 0].bar(alert_counts.index, alert_counts.values, color=colors, edgecolor='black')
    axes[1, 0].set_title('Alert Distribution', fontweight='bold')
```

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```
    axes[1, 0].set_ylabel('Count')
    axes[1, 0].set_xlabel('Alert Type')
    for i, v in enumerate(alert_counts.values):
        axes[1, 0].text(i, v + 2, str(v), ha='center', fontweight='bold')

    # Collection efficiency
    collections_by_hour = daily_data.groupby('hour')
['total_collections'].max().diff().fillna(0)
    axes[1, 1].bar(collections_by_hour.index, collections_by_hour.values, color='purple',
edgecolor='black', alpha=0.7)
    axes[1, 1].set_title('Collections per Hour', fontweight='bold')
    axes[1, 1].set_ylabel('Number of Collections')
    axes[1, 1].set_xlabel('Hour of Day')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/waste_report_project/fig_fill_level_analysis.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_fill_level_analysis.png")
    plt.close()

    # 2. Alert Priority Distribution
    fig, ax = plt.subplots(figsize=(12, 6))

    priority_counts = daily_data['priority'].value_counts()
    priority_order = ['NONE', 'LOW', 'MEDIUM', 'HIGH']
    priority_counts = priority_counts.reindex([p for p in priority_order if p in
priority_counts.index], fill_value=0)

    colors_priority = {'NONE': 'green', 'LOW': 'yellow', 'MEDIUM': 'orange', 'HIGH':
'red'}
    colors = [colors_priority.get(p, 'gray') for p in priority_counts.index]

    ax.bar(priority_counts.index, priority_counts.values, color=colors, edgecolor='black',
width=0.6)
    ax.set_xlabel('Alert Priority', fontweight='bold')
    ax.set_ylabel('Frequency', fontweight='bold')
    ax.set_title('Alert Priority Distribution', fontweight='bold', fontsize=14)
    ax.grid(True, alpha=0.3, axis='y')

    for i, v in enumerate(priority_counts.values):
        ax.text(i, v + 5, str(int(v)), ha='center', fontweight='bold')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/waste_report_project/fig_alert_priority.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_alert_priority.png")
```

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```
    plt.close()

    # 3. Bin Performance Comparison
    fig, ax = plt.subplots(figsize=(12, 6))

    bin_performance = daily_data.groupby('bin_id').agg({
        'fill_level': 'mean',
        'alert_type': lambda x: (x.isin(['critical', 'urgent'])).sum()
    }).rename(columns={'alert_type': 'alerts_triggered'})

    x = np.arange(len(bin_performance))
    width = 0.35

    ax.bar(x - width/2, bin_performance['fill_level'], width, label='Avg Fill Level (%)',
           color='steelblue', edgecolor='black')
    ax2 = ax.twinx()
    ax2.bar(x + width/2, bin_performance['alerts_triggered'], width, label='Alerts
Triggered',
            color='red', alpha=0.7, edgecolor='black')

    ax.set_xlabel('Bin ID', fontweight='bold')
    ax.set_ylabel('Average Fill Level (%)', fontweight='bold', color='steelblue')
    ax2.set_ylabel('Number of Alerts', fontweight='bold', color='red')
    ax.set_title('Bin Performance Comparison', fontweight='bold', fontsize=14)
    ax.set_xticks(x)
    ax.set_xticklabels(bin_performance.index)
    ax.grid(True, alpha=0.3, axis='y')

    lines1, labels1 = ax.get_legend_handles_labels()
    lines2, labels2 = ax2.get_legend_handles_labels()
    ax.legend(lines1 + lines2, labels1 + labels2, loc='upper left')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/waste_report_project/fig_bin_performance.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_bin_performance.png")
    plt.close()

    # 4. Waste Collection Efficiency
    fig, ax = plt.subplots(figsize=(12, 6))

    waste_by_hour = daily_data.groupby('hour')['waste_collected_liters'].sum()

    ax.bar(waste_by_hour.index, waste_by_hour.values, color='orange', edgecolor='black',
alpha=0.7)
    ax.set_xlabel('Hour of Day', fontweight='bold')
```

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```
    ax.set_ylabel('Waste Collected (liters)', fontweight='bold')
    ax.set_title('Waste Collection Efficiency by Hour', fontweight='bold', fontsize=14)
    ax.grid(True, alpha=0.3, axis='y')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/waste_report_project/fig_waste_collection.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_waste_collection.png")
    plt.close()

    # 5. System Architecture Diagram
    fig, ax = plt.subplots(figsize=(12, 8))
    ax.axis('off')

    ax.text(0.5, 0.95, 'Intelligent Waste Management System Architecture',
            ha='center', fontsize=16, fontweight='bold', transform=ax.transAxes)

    components = [
        ('Ultrasonic\nSensors', 0.15, 0.75),
        ('IoT\nCommunication', 0.5, 0.75),
        ('Cloud\nServer', 0.85, 0.75),
        ('Alert\nSystem', 0.5, 0.50),
        ('Collection\nManagement', 0.15, 0.50),
        ('Analytics\nDashboard', 0.85, 0.50),
    ]

    for label, x, y in components:
        bbox = dict(boxstyle='round,pad=0.6', facecolor='lightblue', edgecolor='black',
linewidth=2)
        ax.text(x, y, label, ha='center', va='center', fontsize=11, fontweight='bold',
                transform=ax.transAxes, bbox=bbox)

    connections = [
        ((0.25, 0.75), (0.4, 0.75)),
        ((0.6, 0.75), (0.75, 0.75)),
        ((0.5, 0.65), (0.5, 0.60)),
        ((0.25, 0.65), (0.25, 0.60)),
        ((0.75, 0.65), (0.75, 0.60)),
    ]

    for start, end in connections:
        ax.annotate('', xy=end, xytext=start, transform=ax.transAxes,
                    arrowprops=dict(arrowstyle='->', lw=2, color='black'))

    features_text = 'Key Features:\n• Real-time Fill Level Monitoring\n• Automated
Collection Alerts\n• Route Optimization\n• Cost Reduction\n• Environmental
```

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```
Sustainability'
    ax.text(0.5, 0.20, features_text, ha='center', va='top', fontsize=10,
            transform=ax.transAxes, bbox=dict(boxstyle='round', facecolor='lightyellow',
            edgecolor='black', linewidth=1.5))

    plt.tight_layout()
    plt.savefig('/home/ubuntu/waste_report_project/fig_system_architecture.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_system_architecture.png")
    plt.close()


def main():
    """Main execution function"""
    print("=" * 70)
    print("INTELLIGENT WASTE MANAGEMENT SYSTEM SIMULATION")
    print("=" * 70)

    # Initialize system
    system = WasteManagementSystem()

    # Add bins with different types and locations
    bins_config = {
        'BIN-001': {'type': 'general', 'location': 'Main Entrance'},
        'BIN-002': {'type': 'recyclable', 'location': 'Office Area'},
        'BIN-003': {'type': 'organic', 'location': 'Cafeteria'},
        'BIN-004': {'type': 'general', 'location': 'Parking Area'},
        'BIN-005': {'type': 'hazardous', 'location': 'Laboratory'},
    }

    for bin_id, config in bins_config.items():
        system.add_bin(bin_id, config['type'], config['location'])

    print(f"\nSystem initialized with {len(bins_config)} bins:")
    for bin_id, config in bins_config.items():
        print(f"  - {bin_id}: {config['type']} ({config['location']})")

    # Simulate 7 days
    print("\nSimulating 7 days of operation...")
    all_data = []

    for day in range(7):
        simulation_date = datetime.now().date() - timedelta(days=6-day)
        daily_data = system.simulate_day(simulation_date)
        all_data.append(daily_data)
```

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```
        daily_stats = system.daily_statistics[-1]
        print(f"Day {day+1} ({simulation_date}):
Collections={daily_stats['total_collections']}, "
              f"Waste={daily_stats['total_waste_collected_liters']:.1f}L, "
              f"Avg Fill={daily_stats['avg_fill_level']:.1f}%")

    # Combine all data
    combined_data = pd.concat(all_data, ignore_index=True)

    # Generate visualizations
    print("\nGenerating visualizations...")
    generate_visualizations(system, combined_data)

    # Generate report data
    report = system.get_system_report()

    print("\n" + "=" * 70)
    print("SYSTEM REPORT")
    print("=" * 70)
    print(f"Total Bins: {report['total_bins']}")
    print(f"Bin Types: {', '.join(report['bin_types'])}")
    print(f"Total Waste Collected: {report['total_waste_collected_liters']:.1f} liters")
    print(f"Total Collections: {report['total_collections']}")
    print(f"Average Daily Collections: {report['avg_daily_collections']:.1f}")
    print(f"Average Daily Waste: {report['avg_daily_waste_liters']:.1f} liters")
    print(f"Simulation Period: {report['simulation_days']} days")
    print("=" * 70)

    # Save data to CSV
    combined_data.to_csv('/home/ubuntu/waste_report_project/simulation_data.csv',
index=False)
    print("\nSimulation data saved to: simulation_data.csv")

    # Save report to JSON
    report_json = {
        'system_report': report,
        'bins': list(bins_config.keys()),
        'simulation_date': datetime.now().isoformat(),
        'daily_statistics': system.daily_statistics
    }

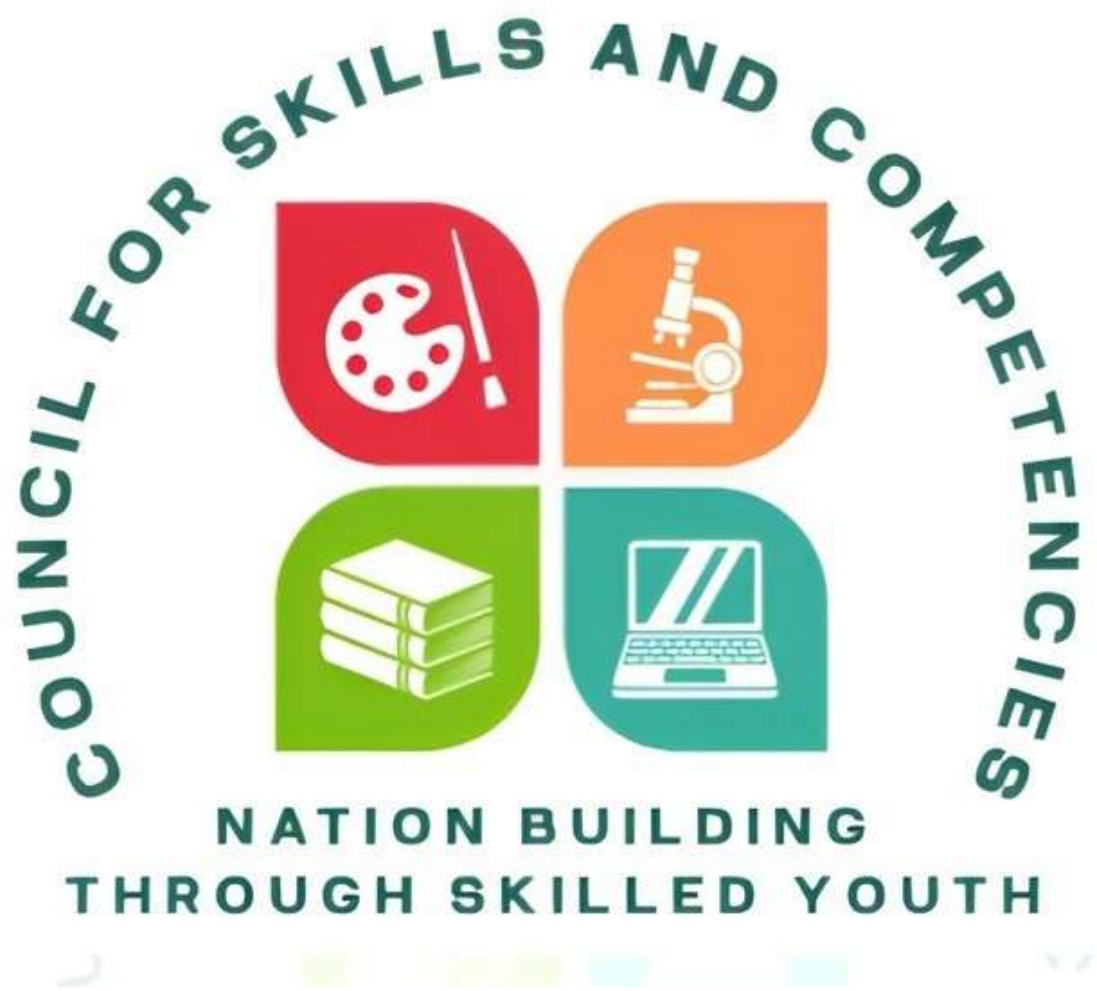
    with open('/home/ubuntu/waste_report_project/system_report.json', 'w') as f:
        json.dump(report_json, f, indent=2, default=str)

    print("System report saved to: system_report.json")
```

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```
    return system, combined_data


if __name__ == "__main__":
    system, data = main()
```



CHAPTER 9

APPENDIX B: SIMULATION DATA SAMPLE

This appendix contains a sample of the raw simulation data generated by the system, demonstrating the granular 30-minute interval logging.

Time	Bin ID	Type	Fill Level (%)	Alert Type	Priority
00:00	BIN-001	general	0.0	normal	NONE
00:00	BIN-002	recyclable	0.0	normal	NONE
00:00	BIN-003	organic	2.0	normal	NONE
00:00	BIN-004	general	1.1	normal	NONE
00:00	BIN-005	hazardous	2.4	normal	NONE
00:30	BIN-001	general	0.0	normal	NONE
00:30	BIN-002	recyclable	4.5	normal	NONE
00:30	BIN-003	organic	2.4	normal	NONE
00:30	BIN-004	general	0.0	normal	NONE
00:30	BIN-005	hazardous	0.7	normal	NONE
01:00	BIN-001	general	1.9	normal	NONE
01:00	BIN-002	recyclable	4.0	normal	NONE
01:00	BIN-003	organic	3.6	normal	NONE
01:00	BIN-004	general	0.0	normal	NONE

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01:00	BIN-005	hazardous	0.0	normal	NONE
01:30	BIN-001	general	0.2	normal	NONE
01:30	BIN-002	recyclable	4.0	normal	NONE
01:30	BIN-003	organic	1.1	normal	NONE
01:30	BIN-004	general	2.0	normal	NONE
01:30	BIN-005	hazardous	0.0	normal	NONE
02:00	BIN-001	general	4.1	normal	NONE
02:00	BIN-002	recyclable	3.9	normal	NONE
02:00	BIN-003	organic	0.8	normal	NONE
02:00	BIN-004	general	0.8	normal	NONE
02:00	BIN-005	hazardous	0.0	normal	NONE
02:30	BIN-001	general	5.9	normal	NONE
02:30	BIN-002	recyclable	7.9	normal	NONE
02:30	BIN-003	organic	0.0	normal	NONE
02:30	BIN-004	general	3.5	normal	NONE
02:30	BIN-005	hazardous	0.0	normal	NONE
03:00	BIN-001	general	1.8	normal	NONE
03:00	BIN-002	recyclable	3.8	normal	NONE
03:00	BIN-003	organic	0.0	normal	NONE

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03:00	BIN-004	general	1.4	normal	NONE
03:00	BIN-005	hazardous	4.8	normal	NONE
03:30	BIN-001	general	0.0	normal	NONE
03:30	BIN-002	recyclable	6.2	normal	NONE
03:30	BIN-003	organic	4.4	normal	NONE
03:30	BIN-004	general	5.2	normal	NONE
03:30	BIN-005	hazardous	0.0	normal	NONE
04:00	BIN-001	general	1.9	normal	NONE
04:00	BIN-002	recyclable	10.6	normal	NONE
04:00	BIN-003	organic	2.9	normal	NONE
04:00	BIN-004	general	4.2	normal	NONE
04:00	BIN-005	hazardous	1.5	normal	NONE
04:30	BIN-001	general	1.9	normal	NONE
04:30	BIN-002	recyclable	12.2	normal	NONE
04:30	BIN-003	organic	0.9	normal	NONE
04:30	BIN-004	general	4.3	normal	NONE
04:30	BIN-005	hazardous	3.1	normal	NONE
05:00	BIN-001	general	5.5	normal	NONE
05:00	BIN-002	recyclable	11.6	normal	NONE

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05:00	BIN-003	organic	5.5	normal	NONE
05:00	BIN-004	general	6.7	normal	NONE
05:00	BIN-005	hazardous	0.0	normal	NONE
05:30	BIN-001	general	3.6	normal	NONE
05:30	BIN-002	recyclable	13.3	normal	NONE
05:30	BIN-003	organic	10.2	normal	NONE
05:30	BIN-004	general	4.3	normal	NONE
05:30	BIN-005	hazardous	0.0	normal	NONE
06:00	BIN-001	general	5.7	normal	NONE
06:00	BIN-002	recyclable	9.5	normal	NONE
06:00	BIN-003	organic	7.8	normal	NONE
06:00	BIN-004	general	3.1	normal	NONE
06:00	BIN-005	hazardous	0.0	normal	NONE
06:30	BIN-001	general	4.7	normal	NONE
06:30	BIN-002	recyclable	13.0	normal	NONE
06:30	BIN-003	organic	3.7	normal	NONE
06:30	BIN-004	general	4.8	normal	NONE
06:30	BIN-005	hazardous	0.3	normal	NONE
07:00	BIN-001	general	5.3	normal	NONE

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07:00	BIN-002	recyclable	10.6	normal	NONE
07:00	BIN-003	organic	8.6	normal	NONE
07:00	BIN-004	general	4.7	normal	NONE
07:00	BIN-005	hazardous	0.5	normal	NONE
07:30	BIN-001	general	11.6	normal	NONE
07:30	BIN-002	recyclable	13.0	normal	NONE
07:30	BIN-003	organic	9.8	normal	NONE
07:30	BIN-004	general	8.2	normal	NONE
07:30	BIN-005	hazardous	1.3	normal	NONE
08:00	BIN-001	general	8.1	normal	NONE
08:00	BIN-002	recyclable	17.1	normal	NONE
08:00	BIN-003	organic	8.7	normal	NONE
08:00	BIN-004	general	7.4	normal	NONE
08:00	BIN-005	hazardous	0.0	normal	NONE
08:30	BIN-001	general	7.2	normal	NONE
08:30	BIN-002	recyclable	12.5	normal	NONE
08:30	BIN-003	organic	10.5	normal	NONE
08:30	BIN-004	general	17.9	normal	NONE
08:30	BIN-005	hazardous	2.9	normal	NONE

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09:00	BIN-001	general	7.6	normal	NONE
09:00	BIN-002	recyclable	15.0	normal	NONE
09:00	BIN-003	organic	10.9	normal	NONE
09:00	BIN-004	general	10.2	normal	NONE
09:00	BIN-005	hazardous	5.7	normal	NONE
09:30	BIN-001	general	11.7	normal	NONE
09:30	BIN-002	recyclable	13.3	normal	NONE
09:30	BIN-003	organic	7.0	normal	NONE
09:30	BIN-004	general	5.6	normal	NONE
09:30	BIN-005	hazardous	2.1	normal	NONE

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CHAPTER 10

APPENDIX C: EXTENDED LITERATURE REVIEW AND METHODOLOGY

Extended discussion on IoT protocols in Smart Waste Management 1. The deployment of IoT sensors in urban environments requires robust and energy-efficient communication protocols. LoRaWAN (Long Range Wide Area Network) is highly suitable for waste management applications due to its long-range capabilities and low power consumption, allowing battery-operated sensors to function for years without maintenance. Cellular networks (NB-IoT, LTE-M) provide higher bandwidth and deeper penetration in dense urban areas, ensuring reliable data transmission from bins located in challenging environments such as underground parking lots or narrow alleys. The selection of the appropriate communication protocol is critical for ensuring the continuous and reliable monitoring of waste bin fill levels.

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